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Machine-Learning Assisted Analysis of Frac Water Hammer

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Abstract

The purpose of this work was to evaluate if frac water hammer signature characteristics are representative of important hydraulic fracture and reservoir properties in a horizontal well and if those characteristics can be used as inputs in a predictive Machine Learning model. Water hammer is an oscillatory pressure signal generated as a result of an abrupt change in wellbore fluid velocity, for example at the end of a fracturing treatment when the pump rate is quickly dropped to zero.

Authors developed an algorithm in python to automatically identify surface pressure and fluid pump rate channels in the raw data, detect and flag end of pumping events, parse out the water hammer and pressure decline signal from raw data. Numerical optimization algorithm was then used to approximate water hammer characteristics from the pressure signal based on the modified damped sine wave equation. The derived equation coefficients were used as inputs (features) in a Random Forest classification model to classify individual fracture stage contributions to a horizontal well's production profile.

Production log tool (PLT) results from 8 horizontal unconventional wells and corresponding 1-second field data for 78 hydraulic stage fracture treatments were used in the study. The water hammer characteristics and parameters of the theoretical vs. actual curve match for each stage, such as initial amplitude of the signal, decay rate, phase angle, angular frequency, number of peaks, etc. were used as features for modeling. The data was split into 70 % –30 % train and test sets. A Random Forest Classifier model was trained on the train set to classify individual fracture stages in a horizontal well as either contributing or non-contributing to production. The model was validated against the test set with overall test classification accuracy of 0.71 and F1-Score of 0.72.

Based on the study the authors conclude that water hammer characteristics derived from surface pressure signal via curve-matching technique can be useful for classification of fracture stage contribution to production in a horizontal well.

Introduction

Water hammer is an oscillatory pressure signal generated by an abrupt change in wellbore fluid velocity - for example at the end of a fracturing treatment when the pump rate is quickly dropped to zero. After frac pumps are shut down, a few minutes of pressure decline, including the water hammer signature, are

routinely recorded on every fracture stage. Operators worldwide collect large amounts of such pressure decline data each year, but these data are rarely used by the operators in the post-stage analysis.

Evaluations from previous papers and the scientific community consider the different wave properties to showcase the elements of the created fracture network and wellbore connectivity (Iriarte et al. 2017, Hwang et al. 2017). Modelling from Carey et al. (2015) first utilized and demonstrated the properties application to the connection of the wellbore. In their work, Carey et al. developed a numerical model for simulating the water hammer signatures. The wellbore with hydraulic fractures was represented by a circuit with resistance (R), capacitance (C), and inertance (I) connected in series. Then, by iteratively changing the values of R, C, and I, they obtained a match between field and simulated data. After a match was obtained, they performed a sensitivity analysis to evaluate the effect of each parameter on the water hammer signal behavior. Based on the analysis of simulations they concluded that water hammer pressure signatures are affected by the created fractures. Iriarte et al. (2017) in their work analyzed a dataset of more than 100 wells in 3 different US plays. They characterized water hammer signatures by calculating decay rate, amplitude, duration, and period. To calculate the decay rate, they obtained a best fit of an exponential decay function to the water hammer peaks (Eq. 1). The process of obtaining the best fit and deriving the decay rate coefficient is summarized in Figure 1. Once the parameters were obtained, they evaluated wellbore, stimulation, and formation effects separately by building correlations between water hammer parameters and the parameters of interest in each of the separate parts. They found that amplitude is closely correlated with the number of perforations per stage, pump rate, and friction, while the duration and decay rate correlate with fluid type, fluid volume, and increased fracture network complexity. The wells and fracture treatments reviewed in our work all had similar treatment designs, fluid types and wellbore construction, which enabled us to eliminate the wellbore and frac fluid parameters and focus on possible formation and fracture network effects on water hammer signal.

$$y(t) = Ae^{-\lambda t} \quad (1)$$

Where A is the amplitude, λ is the decay constant, and t is time.

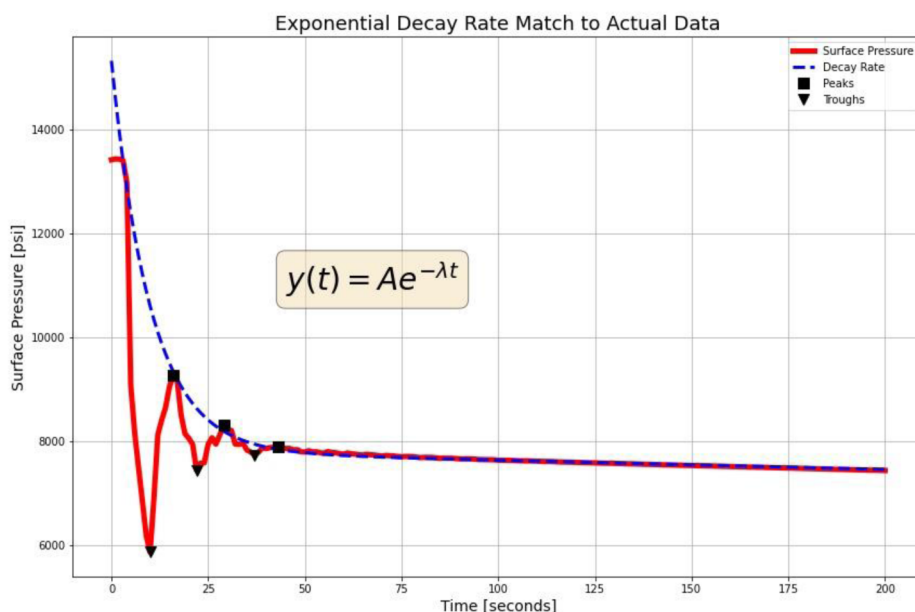


Figure 1—Best fit of an exponential decay function to the water hammer signature peaks with the equation displayed

Methods

Water Hammer Signatures Analysis Workflow

Our study focuses on determining water hammer characteristics by applying an optimization algorithm and obtaining a match between actual field data and a modeled curve based on a modified damped sine equation.

The dataset used in this work consisted of field treatment data for 210 frac stages from 17 horizontal multi-stage wells performed by several frac service providers. A typical stage data file is represented by a tabular format CSV or text file with multiple frac treatment data channels such as surface pressure, pump rate, proppant concentration and other channels, recorded at 1-second intervals.

In the first phase of the analysis, we derived the water hammer signature parameters from the raw field data. Although some stages had multiple pumps shut down events, our study focused on the end of treatment water hammer signature for consistency. Some additional water hammer parameters, such as peaks, troughs and water hammer duration were also taken based on the data during this phase.

To automate the workflow a python script was developed consisting of the following steps:

1. Load the treatment data and identify surface pressure and slurry rate data channels. A significant challenge we faced during this step was inconsistency in data channel naming conventions by different frac service providers. A word stemming/matching technique was used to recognize pressure and rate channels although a more robust machine learning model will probably be required in order to generalize and scale the workflow up.
2. Using a combination of a rule-based algorithm and machine-learning model, identify the end of pumping events and parse out water hammer signature and pressure decline at the end of a frac stage (Figure 2).
3. Identify peaks and troughs (local maximums and minimums) in the data and calculate water hammer signature duration in seconds as time difference between the first and last pressure pulses (Figure 3).
4. Approximate water hammer signature parameters by applying a curve fitting algorithm.

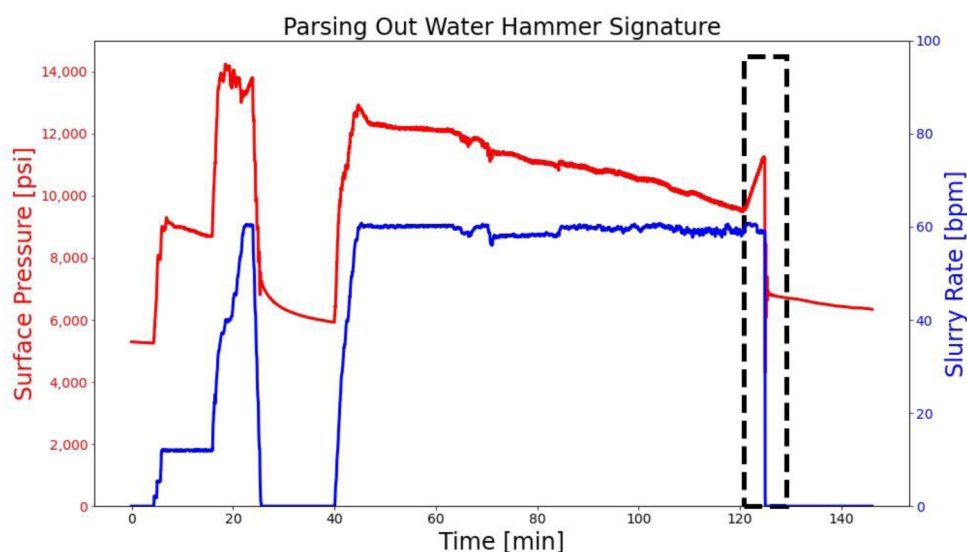


Figure 2—Fracture stage data with end of pumping event and pressure decline including water hammer signature marked

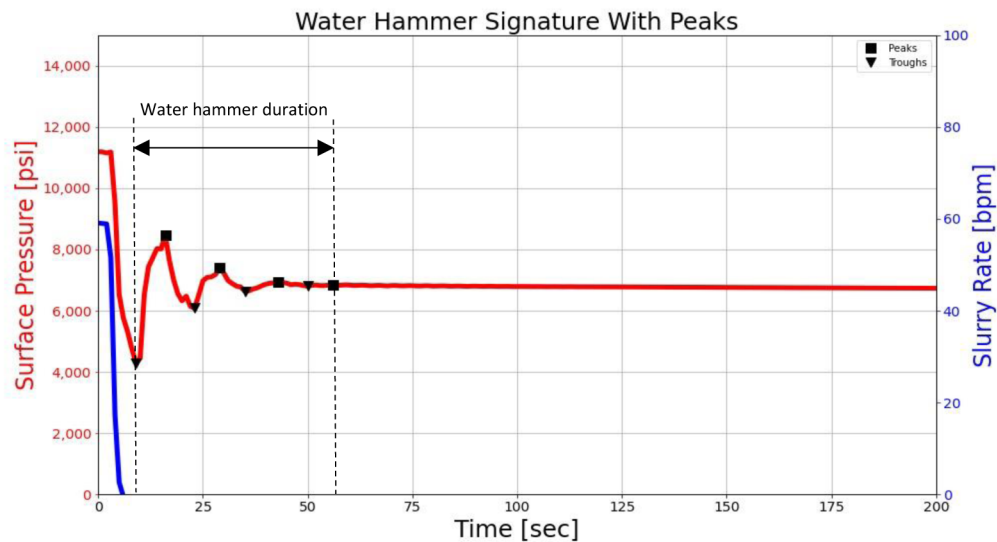


Figure 3—Example of a water hammer signature cutout from data with peaks, troughs and water hammer duration displayed

For the fourth step a curve-fitting algorithm based on numerical optimization was used. As illustrated in Figure 4, the water hammer signatures can vary greatly from stage to stage, even in similar wellbore conditions. One of the objectives of this study was to build a workflow capable of capturing the variations in water hammer pressure decline signatures and learn if these variations can be representative of hydraulic fracture geometry or reservoir parameters.

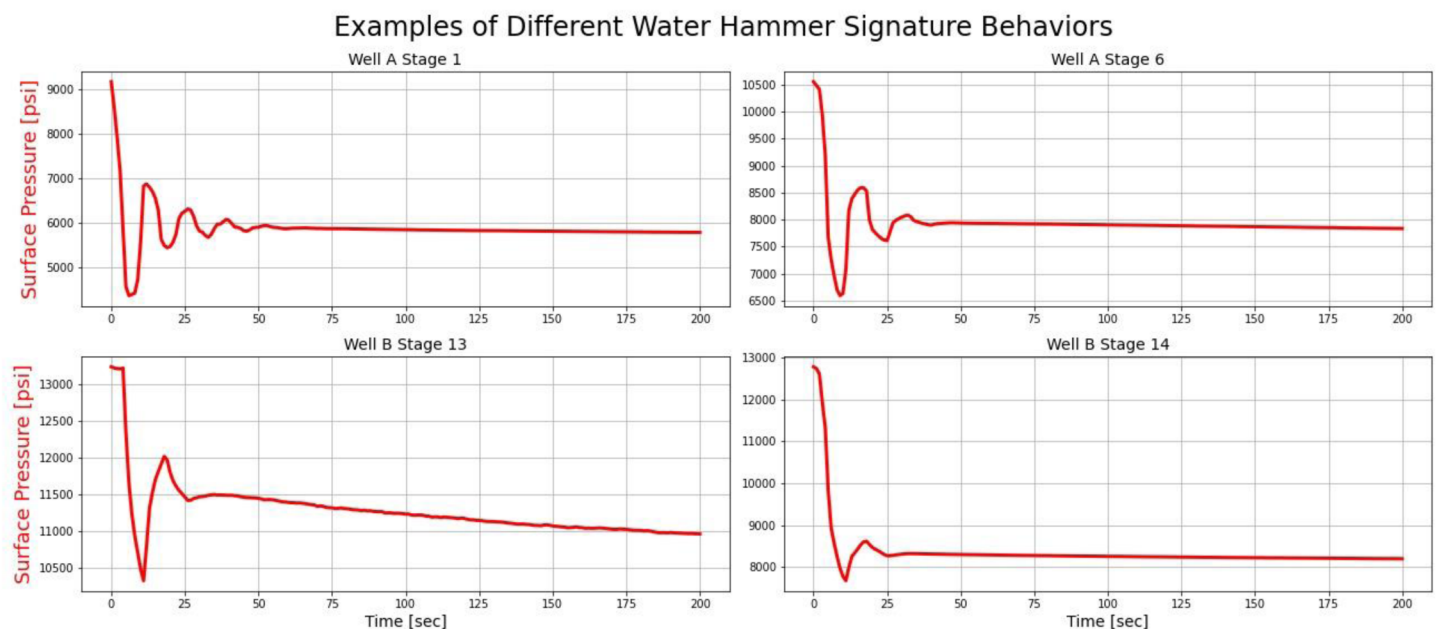


Figure 4—Examples of different water hammer signatures, actual field data from 2 wells / 4 stages.

Overall, water hammer signatures resemble a damped harmonic motion, the type of motion where the energy is lost with each oscillation due to friction. Some examples of the damped harmonic motion are a bouncing ball, a playground swing, a guitar string, etc. This type of motion can be described by a damped sine wave equation (Equation 2).

$$y(t) = Ae^{-\lambda t} \cos(\omega t + \phi) \quad (2)$$

Where A is the amplitude, or half of the initial, the highest, pressure delta between a peak and a trough, λ is the decay constant of the sine wave, ω is the angular frequency or rate of change of the angle of the oscillations and ϕ is the phase angle at $t=0$, used to shift the function to match the initial conditions of the data.

In Figure 5 (a) we show a graph of a damped wave sine function for t between 0 and 10 with some arbitrary function parameters. To account for a pressure decline observed in the real water hammer signature data we added a slope term m to the original equation (Equation 3).

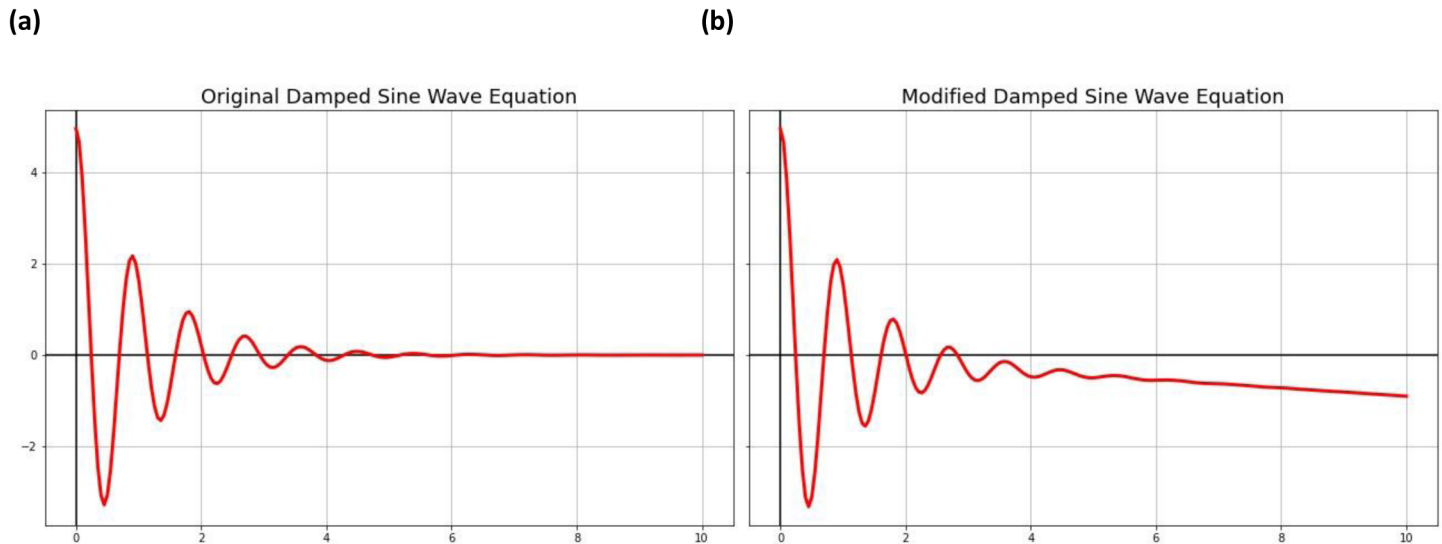


Figure 5—(a) Damped Sine Wave; (b) Damped Sine Wave with added slope.

$$y(t) = Ae^{-\lambda t} \cos(\omega t + \phi) - mt \quad (3)$$

The graph of the modified damped sine wave function is shown in Figure 5 (b)

To illustrate how different water hammer behaviors can be modeled by the damped wave sine equation we graphed the damped wave sine equation for different values of decay constant and angular frequency, while keeping all other parameters constant (Figure 6). As can be seen the higher the decay constant the quicker the energy is lost and therefore fewer peaks are observed with lower amplitude. Higher angular frequency results in higher number of peaks within the same signature duration. Both the decay constant and angular frequency can be useful for matching number of peaks, water hammer duration, and initial amplitude.

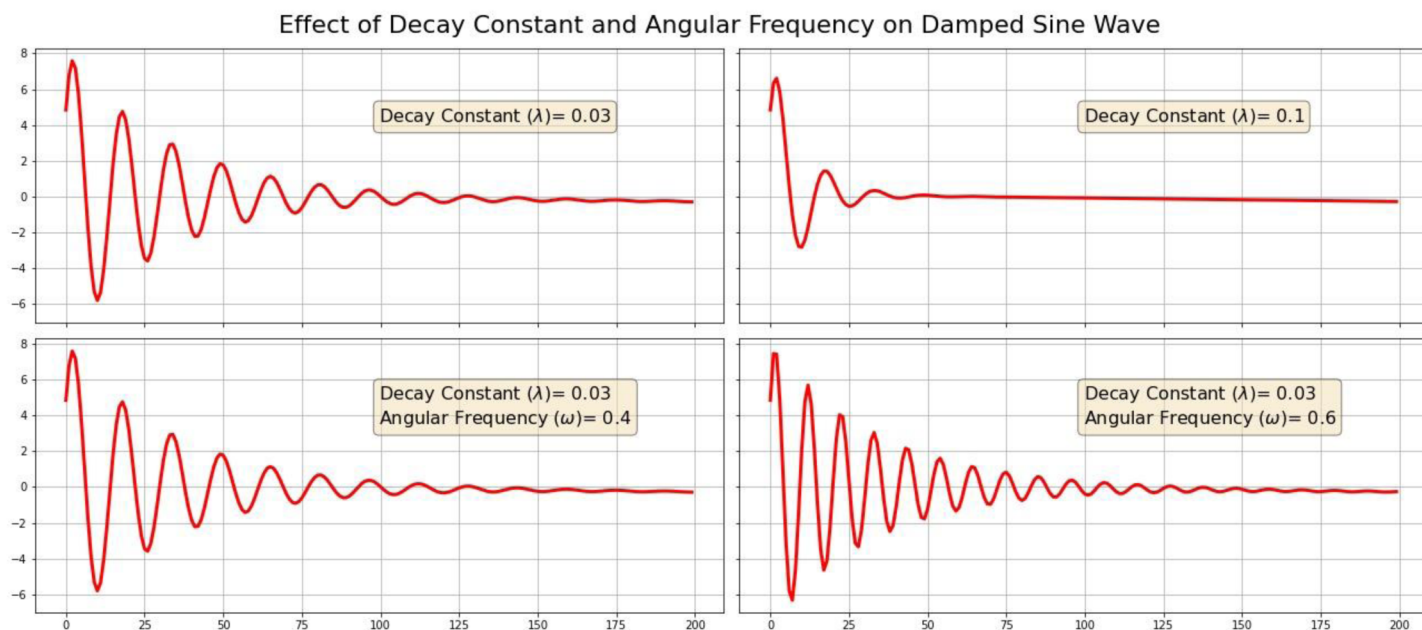


Figure 6—Examples of Decay Constant and Angular Frequency Effect on Damped Sine Wave Behavior

Numerical optimization technique was used to obtain water hammer characteristics from the actual data by applying trust-region least squares algorithm as described by [Branch, Coleman and Li, 1999](#). Surface pressure data was standardized by subtracting mean and dividing by standard deviation before applying the numerical algorithm. By iteratively optimizing the damped sine wave equation parameters (A , λ , ω , ϕ , m) a match was obtained for every frac stage in the dataset and the curve parameters were recorded to create a feature dataset for subsequent use in a machine learning model. Examples of matched actual field water hammer data and modeled by the damped sine wave are provided in [Figure 7](#). The surface pressure data was scaled back to the original scale before graphing.

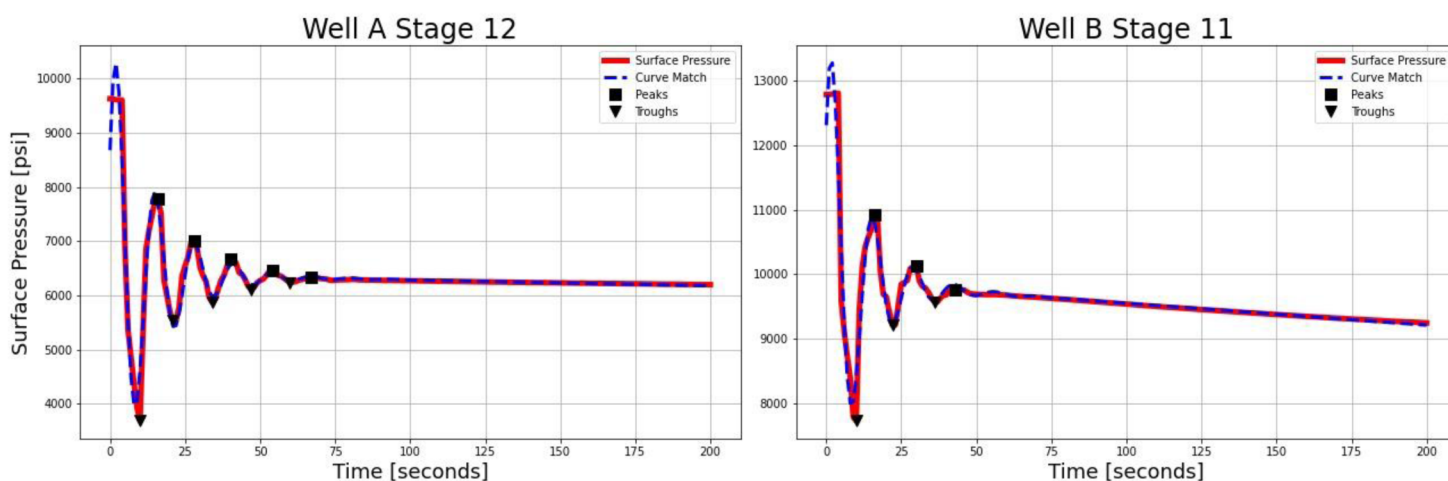


Figure 7—Examples of curve match of Damped Sine Wave and Actual field data, 2 frac stages.

As a result of the first phase of this study a dataset of features was created, and a sample of the dataset with 10 random stages is shown in [Table 1](#). The rows of the dataset represent observations (individual stages) and the columns represent the features or curve characteristics obtained via the curve-matching technique.

Table 1—Water hammer curve parameters derived via curve matching algorithm.

Stage	Number of Peaks	Duration	Amplitude	Decay Constant	Angular Frequency	Phase Angle	Slope
4	4	48	7.257	0.091	0.442	−0.774	0.00186
16	3	32	8.656	0.148	0.436	−1.075	0.00470
1	2	24	7.689	0.114	0.387	−0.838	0.00307
1	3	50	6.909	0.079	0.382	−0.799	0.00062
7	3	34	7.507	0.092	0.455	−0.931	0.00057
19	4	43	8.110	0.107	0.492	−0.947	0.00096
12	5	57	6.805	0.110	0.481	−1.039	0.00161
18	3	33	8.153	0.071	0.465	−1.032	0.00146
6	4	47	9.131	0.171	0.361	−1.026	0.00436
6	4	46	8.556	0.130	0.395	−0.911	0.00124

An Example Application of Water Hammer Signature Analysis Using Machine Learning

In the second part of the study, we investigated if the water hammer characteristic were useful for analysis of oil and gas well production performance. In order to do that we collected Production Logging Tool (PLT) reports from 8 wells, for which we already had the frac stage data and had derived the water hammer characteristics. The PLT reports contained well flow profiles versus measured depth and an interpretation of individual stage contribution to the overall well production for a total of 78 frac stages. The dataset with features created in the first part of the study was then divided into two classes based on individual stage contribution. The stages with contribution less than 5% to the overall production were assigned to "Class 0" or low-contribution stages while the stages contributing more than 5% were assigned to "Class 1" or high-contribution stages. The arbitrary cut-off of 5% was selected for class balancing purposes before using the data in the classification machine learning model. In [Figure 8](#) we show comparison of distributions of different water hammer parameters for the two classes.

Overall, the water hammer signature parameters look similar for both classes. One notable observation is that the distributions of values of amplitude, decay constant and angular frequency look more grouped around the mean for "Class 1" than for "Class 0". Some other observations are higher median angular frequency for Class 1, higher median slope for Class 1, slightly higher median decay constant for Class 1. [Iriarte et al \(2017\)](#) identified a slightly positive correlation between decay rate and 30-day cumulative oil production as wells as a moderate positive correlation between water hammer duration and 30-day cumulative oil production, suggesting that fracture network size may influence the water hammer signal. [Ciezobka \(2016\)](#) discussed that larger fracture networks absorb energy faster therefore leading to faster decays and shorter water hammer duration. While in our study we also noticed a slightly faster water hammer signature decays for high-contribution class, we did not observe significant differences in water hammer durations for the two classes.

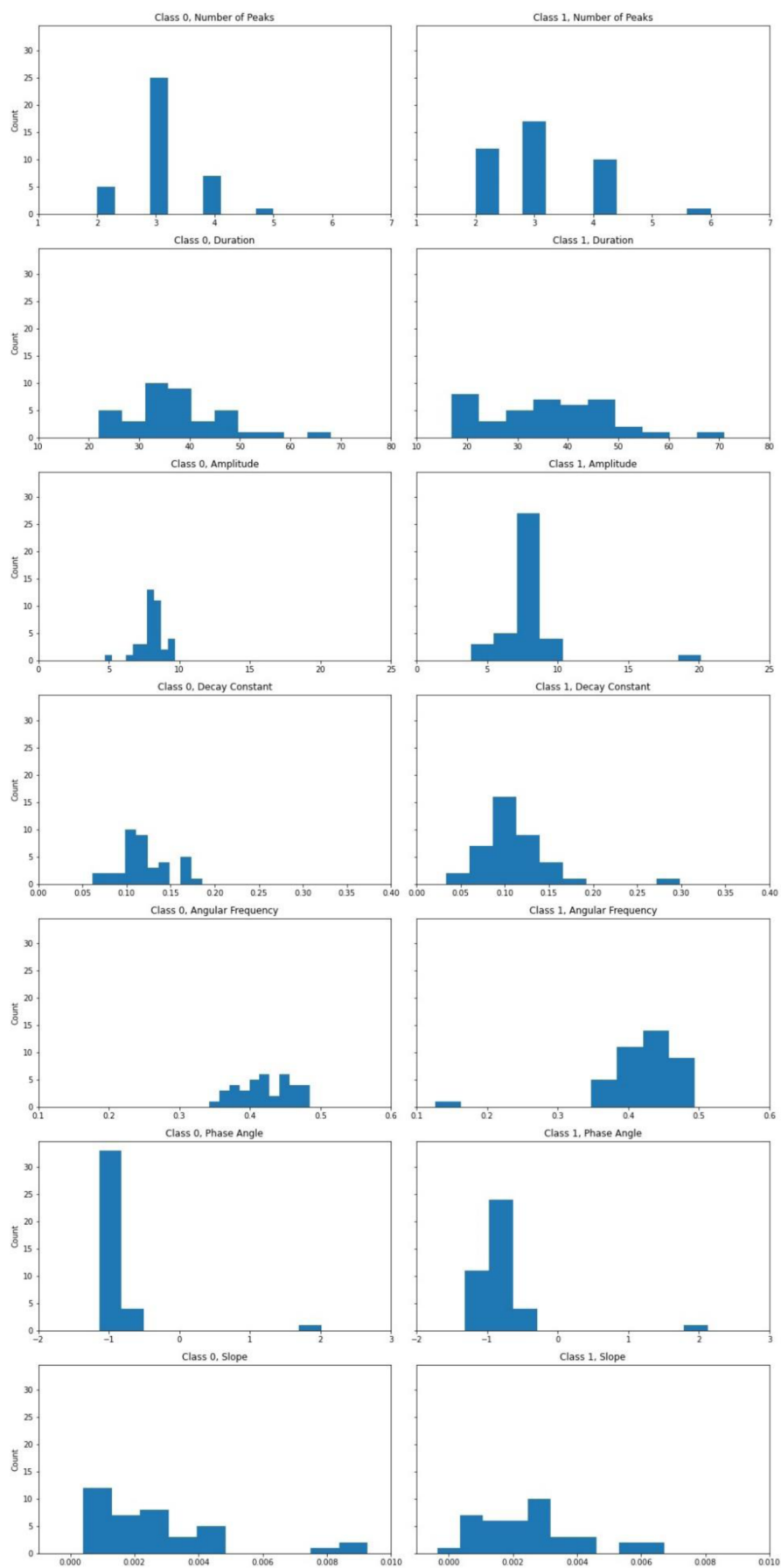


Figure 8—Comparison of water hammer curve parameters for the 2 classes.

In Figure 9 we display a "median" water hammer signatures for both high-contribution and low-contribution stages. To construct these graphs, we calculated medians of each parameter of Equation 2 for both classes, computed pressure as a function of time for 200 seconds and then applied de-standardization transformation. We can see that while the "median" stages have similar water hammer signature, the most obvious difference is the higher slope of the pressure decline for high-contribution class which may be a characteristic of a higher fluid leak-off in the fracture and better wellbore to formation connectivity.

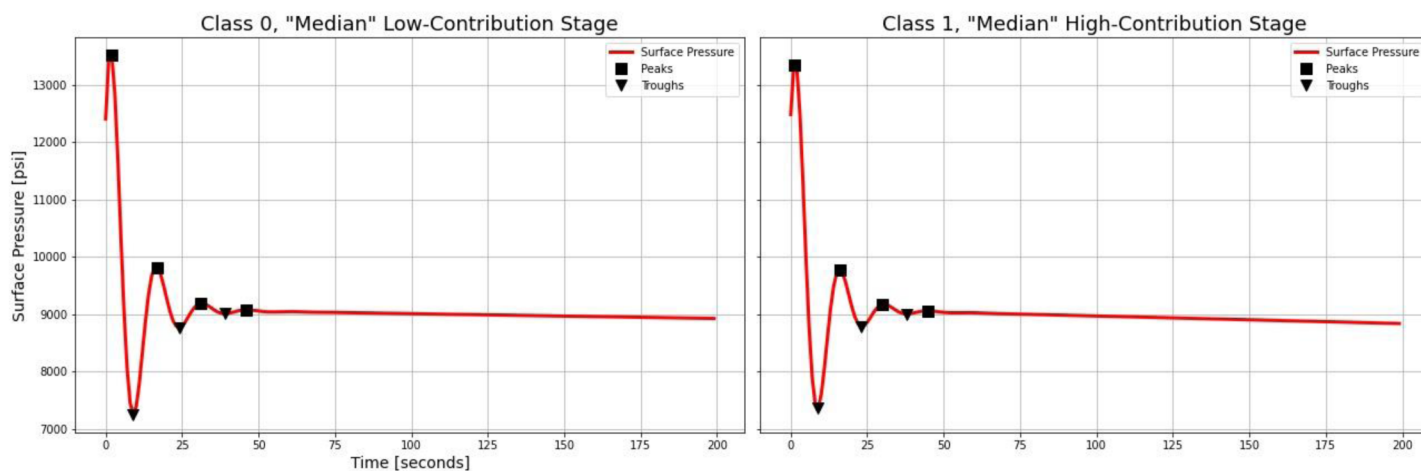


Figure 9—Typical or "median" water hammer signatures for the 2 classes.

Next, a machine learning classification model utilizing Random Forest algorithm was trained to classify stages as either high-contribution or low-contribution stages as a function of water hammer signature parameters. The dataset for this task was split into 70-30% train-test split resulting in 54 observations for training and 24 observations for testing. Once the model was trained to classify stages based on the water hammer characteristics, a prediction was made for every observation in the test set and prediction results were compared to the actual class. In Figure 10 we display the Confusion Matrix calculated based on the test data predictions.

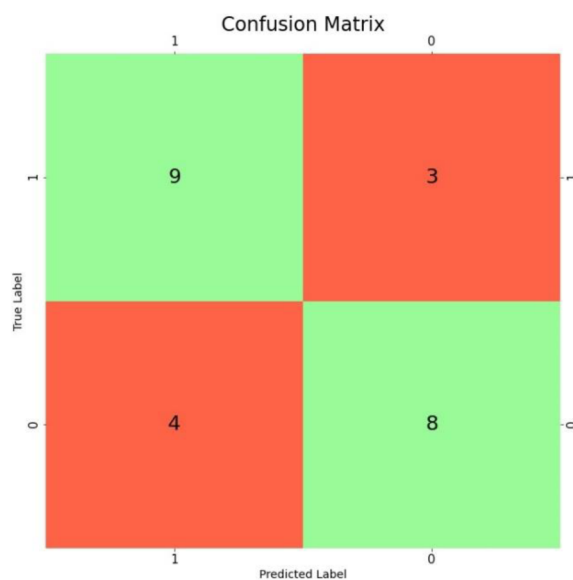


Figure 10—Confusion Matrix for Random Forest model, test data.

The green boxes represent the observations for which the model made correct predictions while the red boxes represent the misclassified observations. If we consider high-contribution class as the positive class and low-contribution class as negative class, then if the model predicted an observation to be positive and the actual class is also positive, such prediction is called a True Positive (TP). Similarly, the correct negative class prediction is called a True Negative (TN). Mis-classified predictions are called a False Positive (FP, predicted as positive but actual is negative) and False Negative (FN). As we can see from the Confusion Matrix, the model correctly classified 9 stages as high-contribution class and 8 stages as low-contribution class while mis-classified 4 and 3 stages respectively. Authors used Overall Accuracy and F1-Score metrics calculated based on the Confusion Matrix to evaluate the model's performance:

$$\text{Overall Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{F1-Score} = \frac{2 * TP}{2 * TP + FP + FN} \quad (4)$$

The resulting accuracy, or the ratio of correctly classified observations, calculated for the test set was 0.708 and the F1-Score was 0.720. Next, we analyzed the relative feature importance as determined by the Random Forest model (Figure 11). The relative feature importance shows which of the water hammer signature parameters are most helpful in predicting if a frac stage will be high- or low- contribution stage. As the model suggests the Slope, Amplitude, and Angular Frequency are the top 3 indicators.

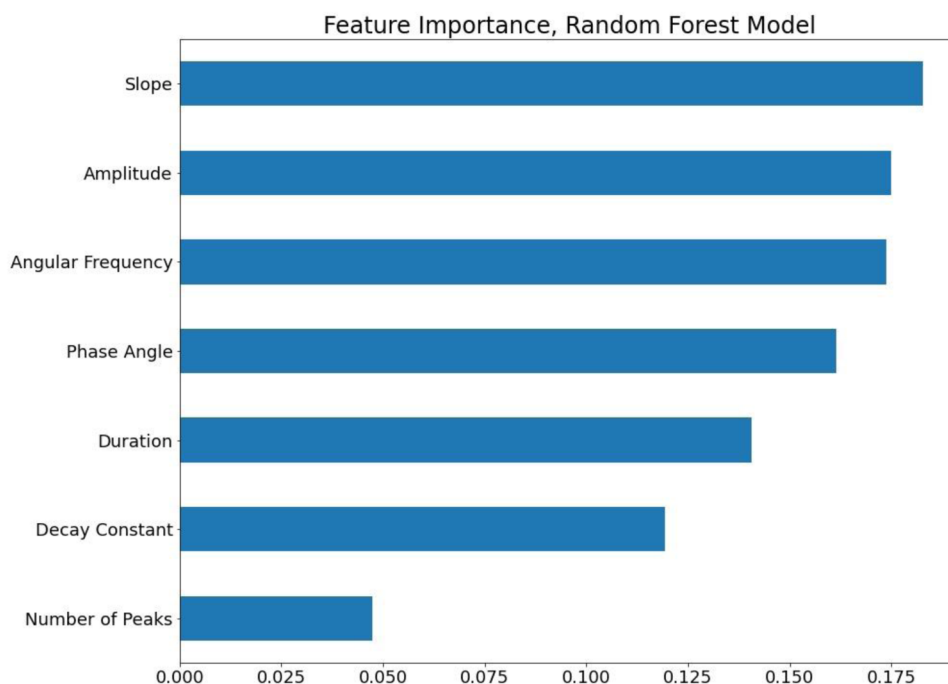


Figure 11—Feature Importance, Random Forest model.

Conclusions and Future Work

Authors developed a workflow to pre-process the raw field frac data, derive water hammer signature characteristics by using a numerical optimization algorithm and then used those characteristics as input parameters for a Machine Learning Model. The model was trained to classify the frac stages as either high-contribution or low-contribution to the overall well production profile based on PLT results. The resulting model accuracy of 0.708 and the F1-Score of 0.720 indicate that the water hammer signature characteristics can be useful in analysis of fracture stage contribution to production in a horizontal well.

Authors intend to collect more field frac stage data, generalize the algorithm to account for differences in fracture treatment design, wellbore construction and frac fluid type, test and re-train the machine learning model, and explore applications of other Machine Learning algorithms such as Artificial Neural Networks and Gradient-Boosting Decision Trees to the task. Some other possible applications of the workflow such as for example detecting the frac plug isolation failures are also considered.

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